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Summary
The article highlights newer aspects of strength training after knee trauma that need to be considered in the future. Recent findings on muscle contraction (three-filament theory) and the importance of eccentric training for prevention and rehabilitation are discussed. The altered central, spinal, and peripheral neuromuscular activation patterns will be addressed after knee trauma and during eccentric muscle actions and their causes. Biomechanical references and psychological (anticipatory) aspects are discussed after knee trauma. In distinction to strength training with healthy persons, concrete hints for strength training and its load normative with persons after knee traumas are given. Finally, newer developments, including improved sensor technology, are presented, and concrete advice is given on optimizing strength training after knee joint injuries and surgery.

Keywords
Knee joint – Strength training – Eccentric muscle actions – sEMG – Biomechanics

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Grundlagen der individuellen Belastungssteuerung im Krafttraining nach Knieverletzungen

Zusammenfassung
Der Beitrag beleuchtet neuere Aspekte des Krafttrainings nach Knietraumen, die es in Zukunft zu berücksichtigen gilt. Thematisiert werden neuere Erkenntnisse zur Muskelkontraktion (Drei-Filament-Theorie) sowie die besondere Bedeutung des exzentrischen Trainings für Prävention und Rehabilitation. Eingegangen wird auf die veränderten zentralen, spinalen und peripheren neuromuskulären Aktivierungsmuster nach Knietraumen und bei

REVIEW / SPECIAL ISSUE

Fundamentals of individual loading for strength training after knee injuries

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Introduction

Strength training after knee injuries, before and after surgeries of the knee joints and the surrounding tissues (e.g., capsules, ligaments, tendons, muscles) has been a natural part of preventing and rehabilitating knee traumas for many years [31]. Strength training serves two purposes when rehabilitating knee injuries: enhancing overall fitness and improving specific sports-related abilities. It is essential to consider the effect of strength training on the injured tissue and the level of strength needed to restore sport-specific skills. Considering the extent of the damage and the particular sport is crucial when engaging in strength training exercises for rehabilitation [19–21].
Parts of musculoskeletal rehabilitation in strength training are general training principles serving as orientations for (all!) athletes and coaches and the organization and systematic structure of training [49].
The training principles and the load normative of strength training have been developed over the past decades primarily through experiential knowledge; only in recent years, it has increasingly developed through training experiments with – mostly healthy – athletes.
Training principles and load normative of strength training have been increasingly applied to subpopulations (Ill or injured individuals), and it is uncertain whether the training principles developed for healthy individuals can be used for strength training after knee trauma in the orthopedic field [11,39] (Table 1).
This paper aims to differentiate the strength training developed from sports and its load parameters and to adapt them to the conditions after knee trauma. For this purpose, recent research on the structure and function of the muscles and newer biochemical aspects of the

exzentrischen Muskelaktionen sowie deren Ursachen. Biomechanische Bezüge werden ebenso wie psychologische (antizipative) Aspekte nach Knietraumen besprochen. In Abgrenzung zum Krafttraining mit gesunden Personen werden konkrete Hinweise zum Krafttraining und deren Belastungsnormative mit Personen nach Knietraumen gegeben. Abschließend werden neuere Entwicklungen, u.a. die verbesserte Sensortechnik, dargestellt und konkrete Hinweise zur Optimierung des Krafttrainings nach Verletzungen und Operationen des Kniegelenkes gegeben.

Schlüsselwörter
Kniegelenk – Krafttraining – Exzentrik – sEMG – Biomechanik

Table 1. Traditional load normatives in strength training.

Training (Load) intensity	Intensity of the individual stimulus (%1RM, %RM; VAS)
Training (Load) duration	Exposure time of a single stimulus or a series of stimuli
Training (Load) density	Time ratio between load and recovery phases
Training (Load) frequency	Number of training sessions in a cycle, e.g., per week
Training (Load) volume	Duration and number [repetitions] of stimuli per training session

activation of the muscles after knee trauma will be considered.

Three filament theory, nervous activation, and eccentric muscle actions

The latest developments in muscle physiology and nervous activation provide more detailed explanations for eccentric muscle actions [18,26,48]. Recent research has demonstrated that eccentric muscle actions demand lower levels of neuromuscular activation to generate equivalent force as other muscle actions, leading to decreased energy expenditure for the same level of force production [29–30,32].

Already in 1999, we could show that after knee trauma, peripheral as well as spinal and central nervous factors are responsible for reduced neuro-muscular activation [15]. Additional researchers corroborated our prior research and built upon its findings. They showed that higher brain centers offer altered activation in different brain regions after knee injuries and surgery. These findings must be considered during rehabilitative strength training focusing on eccentric muscle actions [36–37,44]. Another important reason for increasing the implementation of eccentric training forms is that eccentric strength training significantly reduces the likelihood of injuries [32].

Strength training – Technical capabilities and load on an individual biological matrix

Recent advancements in strength diagnostics and monitoring of sports activities and training enable the evaluation and validation of training principles and load normative effectiveness using miniaturized, cost-effective biomechanical sensors. Optimizing the load monitoring leads to optimizing (strength) training and load control, especially for preventive and rehabilitative purposes [4,7,14,25,27–29,34,50].

Strength training loads act as stimuli on an individual biological matrix, leading to neuro-muscular adaptations based on genetic preconditions and transcripts (see Fig. 1).

The challenges for coaches and therapists lie in the fact that a) athletes react to strength training stimuli in a highly individual manner and b) in injured athletes, additional healing processes are at work that act in a highly individual man-

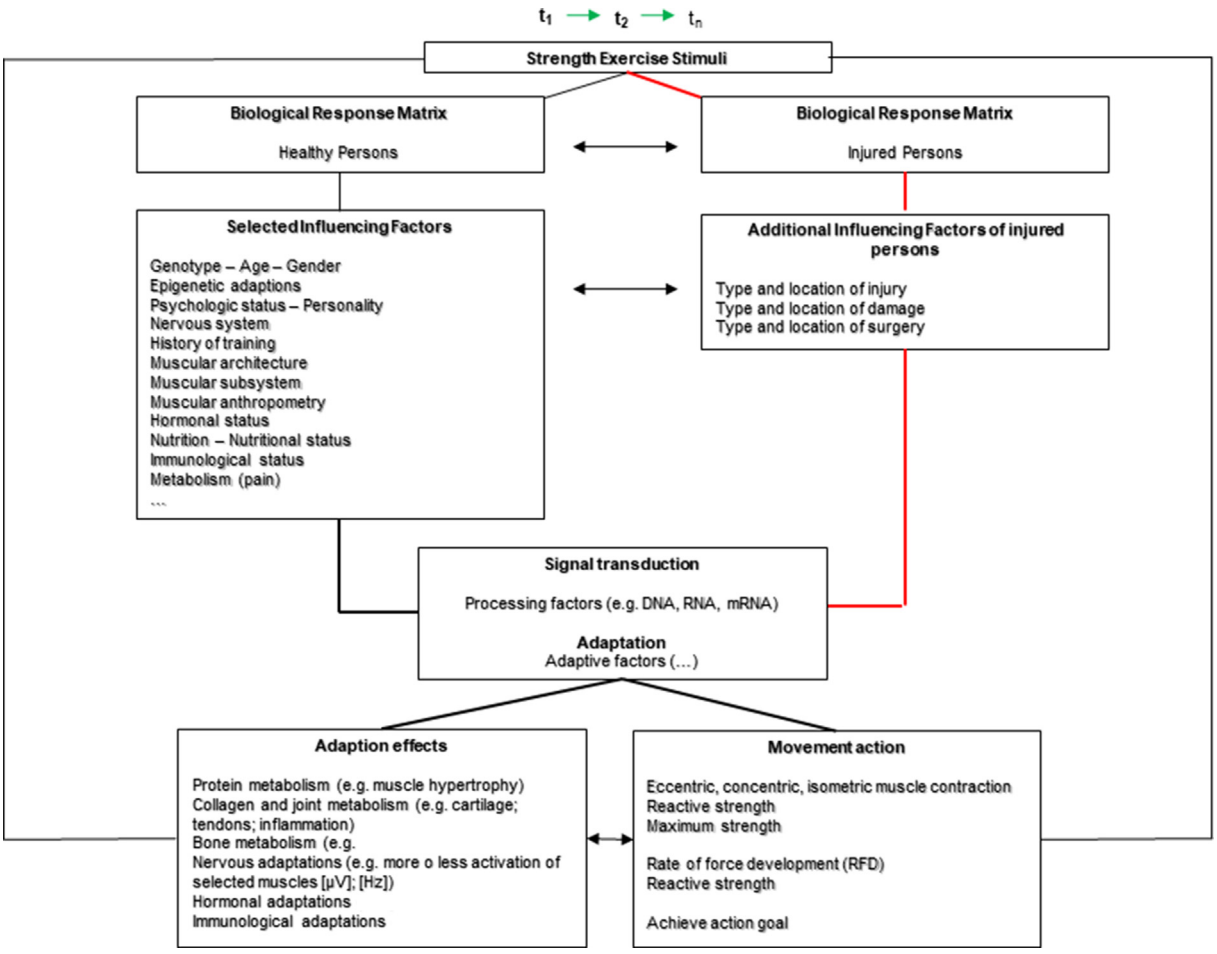

Figure 1
Strength training is a dynamic process with continuous feedback loops. The components of the system change over time, including epigenetic adaptions, ranging from milliseconds (e.g., nervous system; proprioception) to years (e.g., connective tissue; bone).

ner depending on stimulus settings and individual conditions as well as injured tissue types (see Fig. 1).
Fig. 1 displays various factors that impact muscle trainability. Among these factors, the individuality of strength training and the individual responsiveness of each athlete are apparent.

Arthrogenic muscle inhibition, altered central nervous system, and biochemical aspects

Arthrogenic muscle inhibition (AMI) is a limiting factor in muscle strengthening after injuries and surgeries. The muscles around the joint cannot voluntarily be maximally activated, and muscle atrophy occurs even if the patients perform strength training [14]. Less activation of the muscles after injury and surgery depends on central, local, and peripheral factors such as inactivity, post-surgery reduced activities, and healing aspects [42,44–46].
Patients with ACL injuries who show more significant asymmetries and lower performance on the injured side (e.g., hop test) displayed increased activities in partially altered brain regions compared to healthy controls, indicating a higher level of coordination effort and higher energy consumption after knee trauma [6]. Barker et al. (2015) recently found that serum cytokines are enhanced after ACL injuries, corresponding with the strength of the quadriceps muscle [1]. Moreover, joint effusion and swelling of the knee capsule also lead to inhibition of the surrounding muscles, mainly the antigravity muscles [14]. To influence this inhibition process, combining

strength training, anti-inflammatory drugs, and physical therapy such as lymphatic drainage, compression, or ice treatment is essential to reducing joint effusion and inflammation [44,46].

Biochemical reactions can occur from injuries, healing processes or surgery, and strength training [5,40–41]. In rehabilitative strength training, biochemical parameters can be collected, e.g., by blood, urine, and saliva tests. Still, care must be taken to first collect an 'individual baseline' under low load and determine the deflections of the measured parameters under training loads [12].

Various measurements can be taken to establish a reliable baseline for training adaptations, including cortisol/testosterone ratios, adrenaline, noradrenaline levels, Hemoglobin, CK, and urea levels. Deficiency symptoms such as ferritin and iron deficiency (and, in rare cases, magnesium deficiency) should also be checked. Monitoring immune status indicators such as free radicals and antioxidants is also recommended. By establishing a solid baseline, any deviations can be accurately interpreted about training adaptation [12].

The biochemical parameters can provide valuable insights into an individual's load, nutritional, and resorption behavior. They are instrumental in assessing the status of essential nutrients such as vitamins, minerals, and trace elements. By analyzing these parameters, one can better understand an individual's overall health and well-being, training load, and the effectiveness of strength training [3].

Selected aspects of rehabilitative strength training with injured persons

Knee trauma usually impairs the maximum force and the rate of force development (RFD), a critical factor in developing explosive power for high-speed sports. On the one hand, explosive and maximally powerful loads are necessary for sport-specific strength training; on the other hand, they are often impossible with injured people due to damage [39].

Exercise selection is based on the individual, injury-related biomechanical load capacity and the sport-specific requirements. Depending on the injury and the goals (sport-specific), primary objectives include regaining maximum strength, force increase slope, reactive strength, and, in particular, coordinative control concerning a target sport [39]. It should be emphasized that rehabilitative strength training always includes sensorimotor training (coordination) and, therefore, cannot be considered and planned in isolation as a conditional skill [49].

After analyzing EMG data, we found that knee injuries affect coordination and require muscle amplitudes for movement throughout the body [2,14].

> After an injury, it's essential to exercise and train complex movements, not just individual muscles, to achieve the same goals with altered coordination.

Reestablishing muscle activation amplitudes (IV) is comparatively simpler than restoring high-frequency activation [Hz] necessary for activating fast twitch muscle fibers (FTx). Nevertheless, both peripheral and central nervous activation and control of the muscles can be impacted. Severe knee injuries can diminish the CNS's explosive, high-frequency activation of muscles, which may not be fully recoverable [2].

Therefore, not only maximum force is impaired after knee trauma, but especially the rate of force development (RFD), which is one of the essential parameters in explosive force development in high-speed sports.

To effectively activate the FTx fibers with explosive power, it is recommended to utilize high resistance (weights) and maximum acceleration or heavy eccentric load training [16]. Both forms of exercise are restricted to rehabilitative strength training of the injured limb to prevent further damage from high biomechanical loads. As a result, low intensities and high volumes are often used during training sessions. This approach affects the injured limb and the entire body, decreasing neuromuscular activation from the central nervous system [2].

Exercise selection is, therefore, based on the individual, injury-related biomechanical load capacity and the sport-specific requirements. This problem is evident in all sports disciplines. On the one hand, explosive and maximally powerful loads are necessary for sport-specific strength training; on the other hand, they are often impossible due to damage. When choosing exercise, biomechanical considerations must be made.

One example is the biomechanics of the deep knee squat. Deep squats are essential for weight lifters but lead to the highest biomechanical loads on the knee joint (Cartilage, Cruciate ligaments, Menisci, Tendons).
Deep squats are prepared for years in weightlifting, first without and later with increasing weight load.
In most sports, deep squats are not necessary for training-specific joint-angle-strength gains. This makes them unsuitable for exercise, particularly during and after rehabilitation.

Psychological and anticipating aspects

Many factors influence whether an individual will return to sport after this surgery. They include surgical details, rehabilitation, and social and psychological factors [8–9]. After knee trauma, training interventions should be tailored to the individual sensorimotor and psychological context, which influences the planning and execution of movement actions [17,33], as each trauma changes the subjective self-perception and, therefore, affects the planning, execution, and evaluation of movement actions [23–24,47] (Fig. 2).
The anticipation of pain, fear, or insecurity can result in reduced coordination, force production, and explosive power, as well as a decreased rate of force development [13] (Fig. 2).
Suppose an athlete is expecting to experience discomfort while undergoing a maximal strength test, which could result in a decrease in their exertion. The conventional strength training methods can pose a challenge as the therapist relies on the maximum strength reading to devise the training plan without considering the possible impact of pain, fear, or insecurity on the outcome.

Conclusion and recommendations for future rehabilitative strength training after knee injuries

In strength training for rehabilitation, it's essential to consider the latest research on muscle physiology, eccentric training, load characteristics, new technology, genetic adaptation, and psychological factors. Applying evidence-based medicine and training to enhance rehabilitative strength training is crucial to optimizing strength training after knee injuries [10,43]. In the future, the usual load standards must be adapted to the unique requirements of rehabilitative strength training (Table 2).

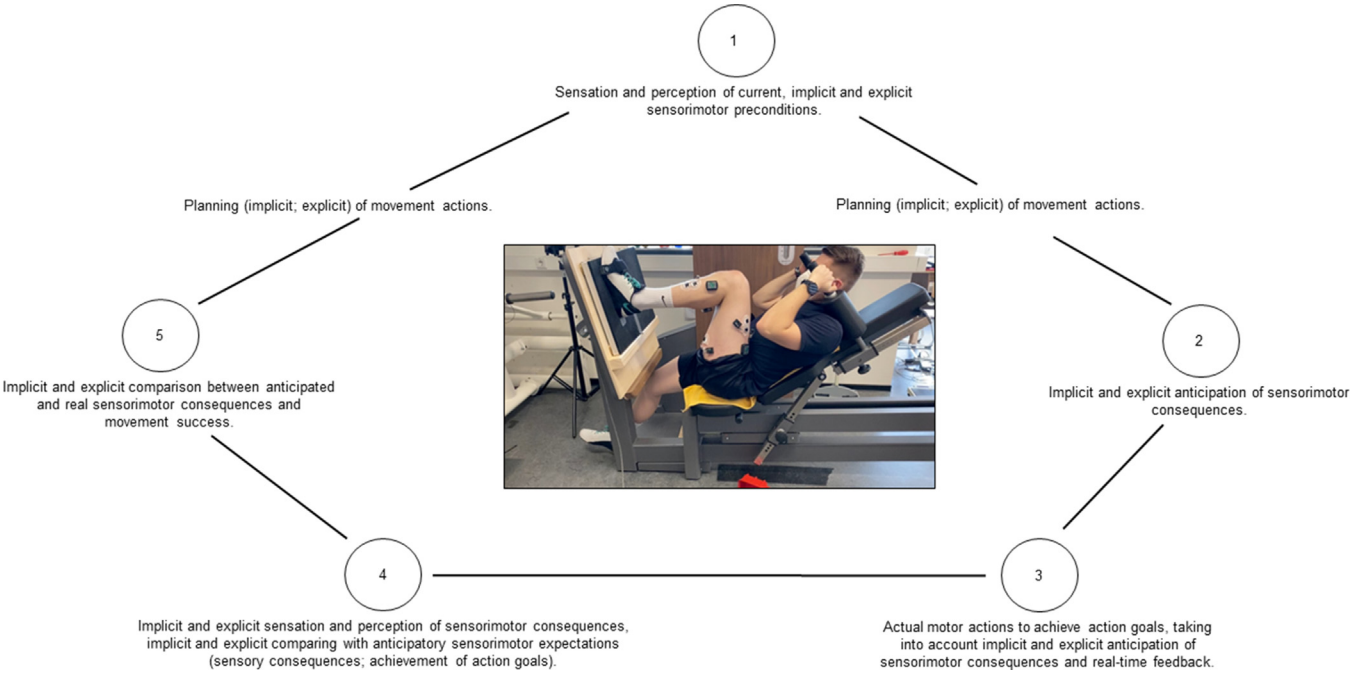

Figure 2
Implicit and explicit planning, planning, and execution of movement actions.

It is recommended to include sensor-based monitoring in rehabilitation, such as GPS, LPS, force transducer, force plates, sEMG, and electronic goniometer, to improve training control and scheduling. Rehabilitative strength training should be more personalized and, in exceptional cases, involve genetic analysis. In high-performance sports, even in rehabilitation, monitoring biochemical levels to regulate physical strain and prevent overtraining may be beneficial, and athletes should receive psychological support to reduce anxiety, uncertainty and enhance their motivation. Including more eccentric strength training after knee trauma is vital when creating training schedules. Scientific studies have shown that eccentric exercises reduce the risk of injury, especially to the hamstring muscles and ACL [35]. Rehabilitative strength training should prioritize complex movements that mimic everyday activities and sports rather than solely targeting individual muscles. The muscles of the non-injured limbs and trunk should be trained more often with explosive force development, as

Table 2. Extended load normatives in strength training after injuries.

Parameter	Description/ Units	General Comments	Special comments for orthopedic rehabilitation
Load intensity	[1RM%]; [MRM%] [Subjective Scales 0-10]	For healthy and trained athletes, the single-repetition maximum is the first choice.	Injured athletes should not measure their single-repetition maximum due to pain and risk of further injury. As a result, maximum force measurement is not a valid or reasonable option.
Load volume	Number of repetitions [n] Duration of repetitions [s], [min]	For healthy and trained athletes, an important parameter to consider is the alignment of load intensity.	It is important to gradually increase exposure and monitor for potential stress reactions such as joint pain, hyperthermia, swelling, and joint effusion.
Numbers of sets	[n]	Number of sets; number and duration of neuromuscular activations	The frequency of exercise should gradually increase, taking into account possible negative reactions such as muscle soreness, joint pain, hyperthermia, swelling, and joint effusion.
Time under Tension	[s], [min]	The recommended time under tension for hypertrophy is 90-120 seconds.	Due to the lower activability after knee trauma (especially of the knee extensors), a longer duration of activation should be aimed for after knee trauma
Type of load - muscle action	Concentric, eccentric, isometric, auxotonic, isoinertial and isokinetic	1st movement phase = moving weight 2nd movement phase = reversing movement 3rd movement phase = return weight	In orthopedic rehabilitation, it is important to practice the alternation between concentric and eccentric muscle action (movement reversal) and to improve neuromuscular control during eccentric movement.
Accentuated metabolic range	ATP; KP Anaerobic-lactacid	Accentuated metabolic range	Depending on the requirements of the sports, different metabolic ranges can be targeted, particularly anaerobic alactacid metabolism (ATP; CK) and lactate metabolism (anaerobic glycolysis).

Table 2 (*continued*)

Parameter	Description/ Units	General Comments	Special comments for orthopedic rehabilitation
Acceleration (Rate of force development, RFD) Load Velocity	Force increase slope (RFD_{max}) [N/s], [m/s^2] [m/s]	To achieve a higher movement speed, the acceleration must be increased. There are different types (Parameter) of RFD: • Average RFD or IES (Index of Explosiveness) • Time-Interval RFD • Instantaneous RFD • Peak or maximal RFD • Time to peak RFD	The advantage of performing strength exercises with low intensity, slow speed, and minimal weight is the reduced pressure on the healing tissue. However, the downside is the activation of predominantly slow-twitch fibers. The increase in force (acceleration of masses/weights) is directly inversely related to the mechanical stress on muscles, tendons, and cartilage. Therefore, at the beginning of rehabilitation, the acceleration of resistance should be kept low. Slow accelerations recruit predominantly ST and FTA muscle fibers. Both when training on equipment and with free weights, attention must be paid to optimal technique, regardless of the speed of movement
Movement reversal points	[m/s^2]	Holding - slow - fast - maximum fast	Different accelerations during movement reversal led to various neuromuscular recruitments and biomechanical loads.
Range of motion (ROM)	[°]	The range of motion influences the time under tension at the same speed of motion.	Damaged or low-load joint areas can be left out by restricting certain joint angles. The use of the entire ROM in strength exercises can increase joint mobility.
Load duration (1) - single contractions	[s]	Time under Tension of one single contraction [s]	The frequency of exercise should be increased slowly, and possible adverse stress reactions should be considered (muscle soreness, joint pain, hyperthermia, swelling, joint effusion).
Load duration (2) - repeated contractions (set; training unit)	[s]	Time under Tension during repeated muscle actions (Total time [s])	The frequency of exercise should be increased slowly, and possible adverse stress reactions should be considered (muscle soreness, joint pain, hyperthermia, swelling, joint effusion).

Table 2 (*continued*)

Parameter	Description/ Units	General Comments	Special comments for orthopedic rehabilitation
Load density Load frequency	[s], [min]	Recovery times between sets vary based on training status and training target.	The duration of rest periods between sets in strength training sessions is determined by various factors, such as one's current load capacity, training condition, and objectives. Recovery times might need to be prolonged when working with orthopedic patients due to the healing processes and adjustments to strength training. It is crucial to ascertain individual recovery times based on their specific load capacity.
Recovery times between training sessions	[n/hours] [n/day] [n/week]	The recovery time between training sessions varies for different levels of training. Untrained people need about 72 to 84 hours, strength-trained athletes about 48 hours, and highly trained athletes 24 to 48 hours.	When training orthopedic patients for strength, it is crucial to tailor the frequency of the sessions to their current load capacity. This is because their healing processes and adaptations to the training stimuli are parallel. Additionally, recovery times between training sessions may be prolonged.
Termination criterions	[n], [s], [min] VAS Protocols Subjective perceptions and protocols.	Reaching the specified number of repetitions [n] or contraction duration [s], Failure to achieve the predetermined range of motion [^{0}], Exhaustion - Loss of technique, Point of Momentary Muscular Failure (PMF). Achievement of PMF is extremely dependent on patient motivation and perception of pain.	When strength training with orthopedic patients, special attention must be paid to the technique of movement execution and whether the patients experience pain. During the initial phases of rehabilitative strength training, training up to PMF is not recommended due to the high biomechanical load and loss of technique. It is crucial to maintain a written record of how orthopedic patients perceive strength training and reasons for discontinuation, if necessary.

maintaining high-frequency activation of the muscles by the central nervous system is of great importance [22,38].

Attention must be focused on reducing atherogenic and central nervous inhibitory processes related to the injured joint. This combines strength training with physical therapy focusing on lymphatic drainage, compression, or ice treatment therapies.

CRediT authorship contribution statement

Jürgen Freiwald: Conceptualization, Investigation, Writing – original draft, Visualization, Supervision. **Sasha Javanmardi:** Investigation, Visualization, Writing – review & editing. **Thilo Hotfiel:** Investigation, Writing – review & editing. **Martin Engelhardt:** Investigation, Writing – review & editing. **Ludwig Rappelt:** Writing – review & editing. **Christian Baumgart:** Conceptualization,

Investigation, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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